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From Chaos to Clarity: AI-Driven Network Troubleshooting

Instructions for WS:D4 workshop participants

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Before we begin

We strongly encourage you to join our Slack channel for this workshop:

- URL: **`https://networkautomationfrm.slack.com`**
- Channel: **`#ac4-ws-d4`**

Get familiar with the setup

The setup used during the workshop consists of two main components:

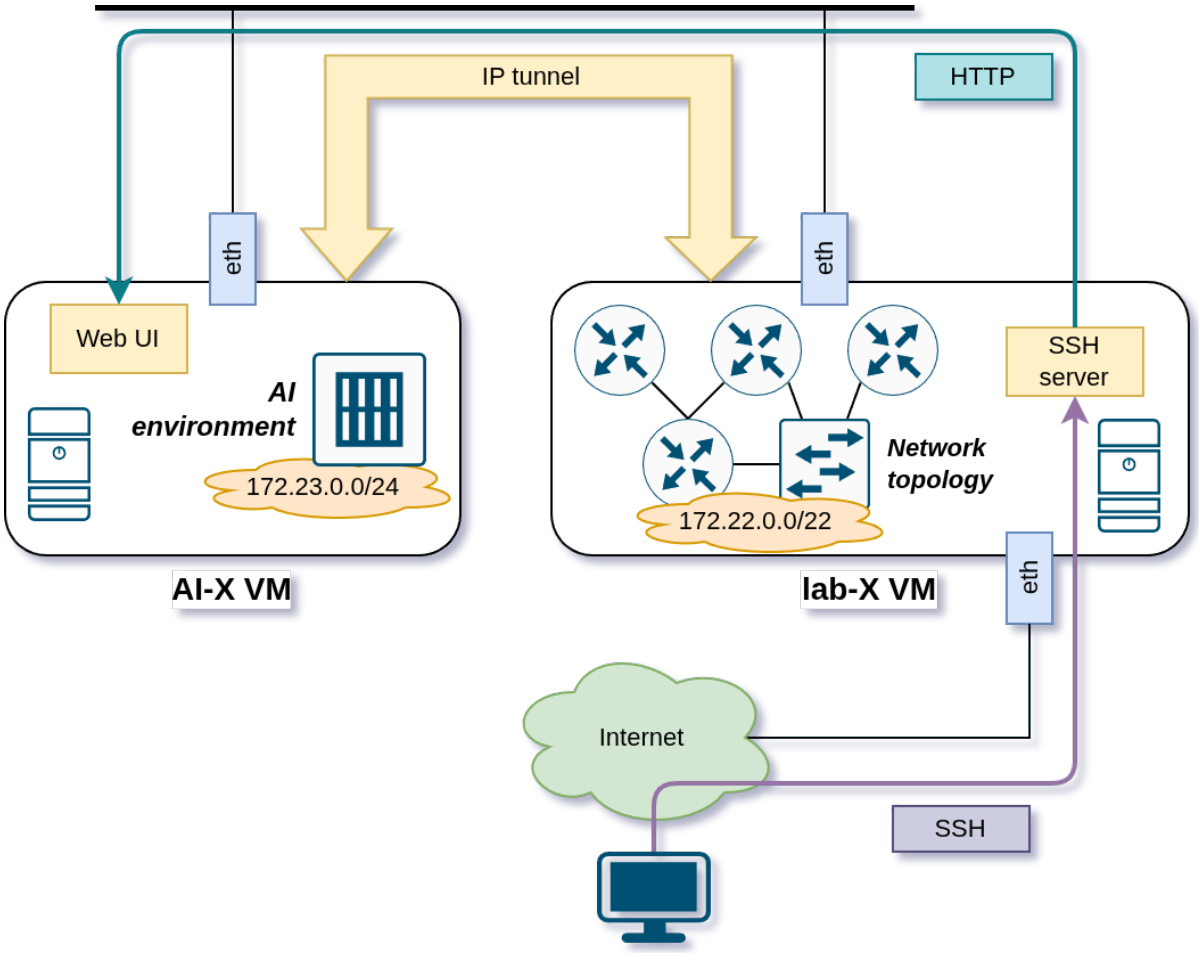
- The AI components running as a dockers on AI VM - no access to VM, access via UI only
- Containerlab environment simulating network topology running on LAB VM - full access to emulated devices

Access to VMs

Each workshop participant will have access to a dedicated lab VM in the cloud. Hostname and login credentials for that VM will be given during the workshop. Access is provided via SSH protocol.

Note: Although the Web UI is hosted on AI VM (with no access for attendees), an IP tunnel between LAB VM and AI VM will be provided for easier access:

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To access Web UI from your local machine, a SSH tunnel is needed: towards 172.23.0.80 IP and 8501 TCP port. More details are provided in the next section.

SSH session with port forwarding using PuTTY

Here's how to set it up in PuTTY (vm_nb and password will be given to you at the beginning of the workshop) :

1. **Open PuTTY**: Launch the PuTTY application.
2. **Session Settings**:

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- In the left pane, go to **Session**.
- Enter the **Host Name (or IP address)** as `lab-<vm_nb>.ac4.codilime.com`.
- Set the **Port** to `22` (default for SSH).
- Under **Connection type**, select **SSH**.

3. **Define the SSH Tunnel (Port Forwarding)**:
   - In the left pane, expand **Connection** > **SSH** and then select **Tunnels**.
   - In the **Source port** field, enter `8501` (the local port).
   - In the **Destination** field, enter `172.23.0.80:8501` (the address and port on the remote server).
   - Ensure **Local** is selected (for local port forwarding).
   - Click **Add**. The forwarded port should now appear in the **Forwarded ports** list as `L8501 172.23.0.80:8501`.
4. **Save the Session** (optional):
   - Return to the **Session** category at the top of the left pane.
   - Under **Saved Sessions**, type a name (e.g., `VM SSH Tunnel`) and click **Save** to easily reuse this configuration in the future.
5. **Connect**:
   - Click **Open** to start the SSH connection.
   - A terminal window will open, prompting you to enter the password for `codi` user.

SSH session with port forwarding using Linux terminal (alternative)

Here's how to set it up in Linux (vm_nb and password will be given to you at the beginning of the workshop) :

1. Open a terminal and type the following command:

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```
# ssh -L 8501:172.23.0.80:8501 codi@lab-<vm_nb>.ac4.codilime.com
```

2. Enter the password for codi user for your VM

Note: Please always keep at least one SSH session open to ensure that the web UI is accessible at all times.

UI overview

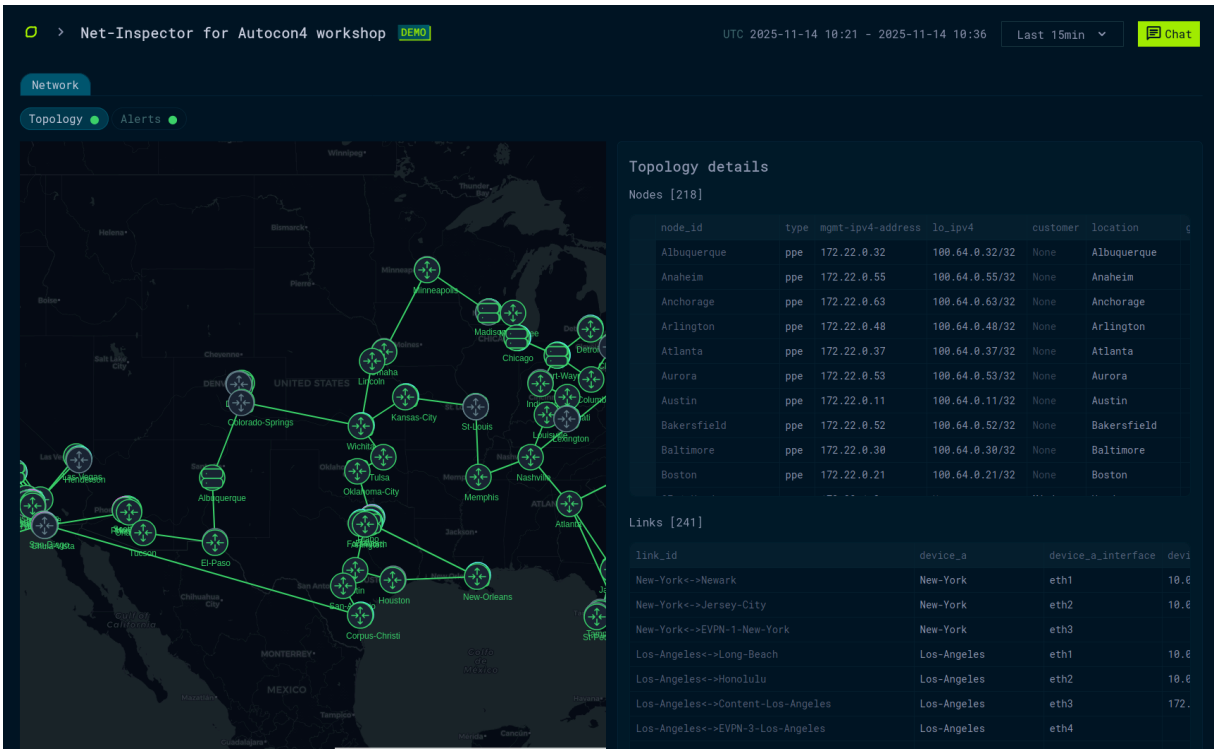
Having SSH tunneling up and running, you can access the web interface of the Net-Chat Assistant:

1. Open your browser (Chrome/Firefox) and paste the link:

   ```
   http://localhost:8501
   ```

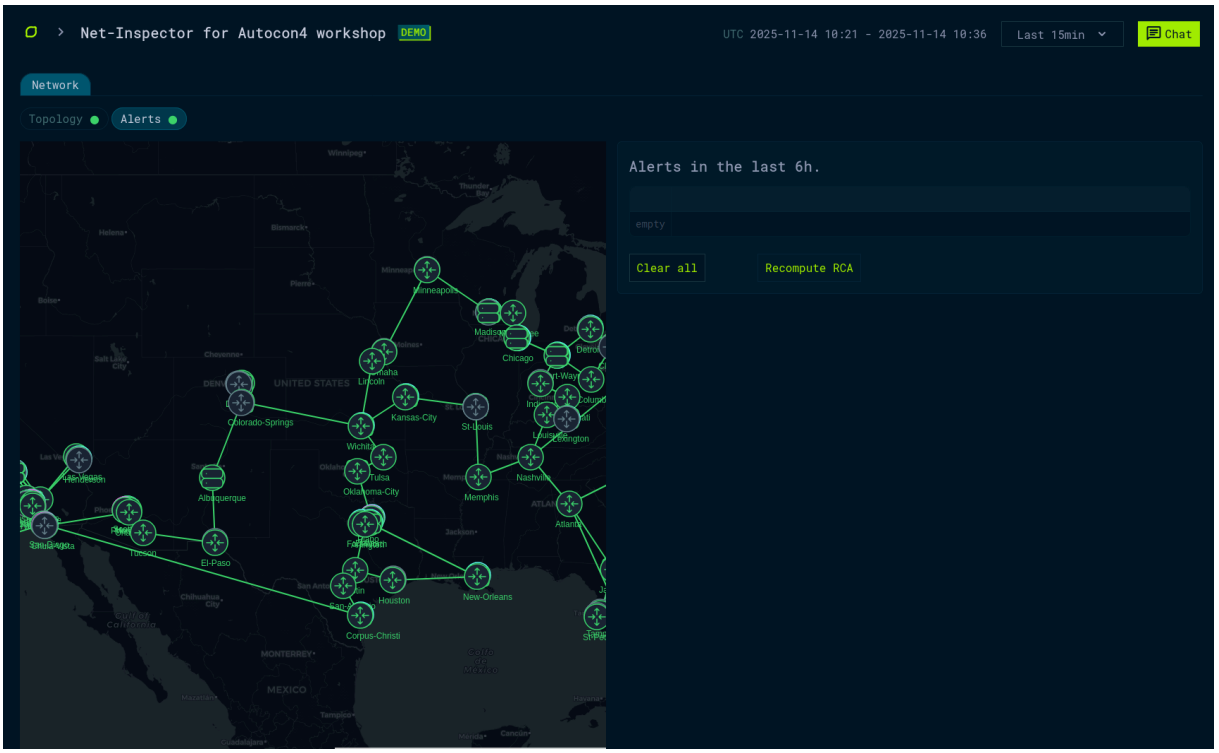
2. You should see the UI of the web frontend.

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3. For now Alerts will be empty and discussed later on during second scenario:

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4. Clicking green “Chat” button one can open chat in order to ask a question regarding topology such as:

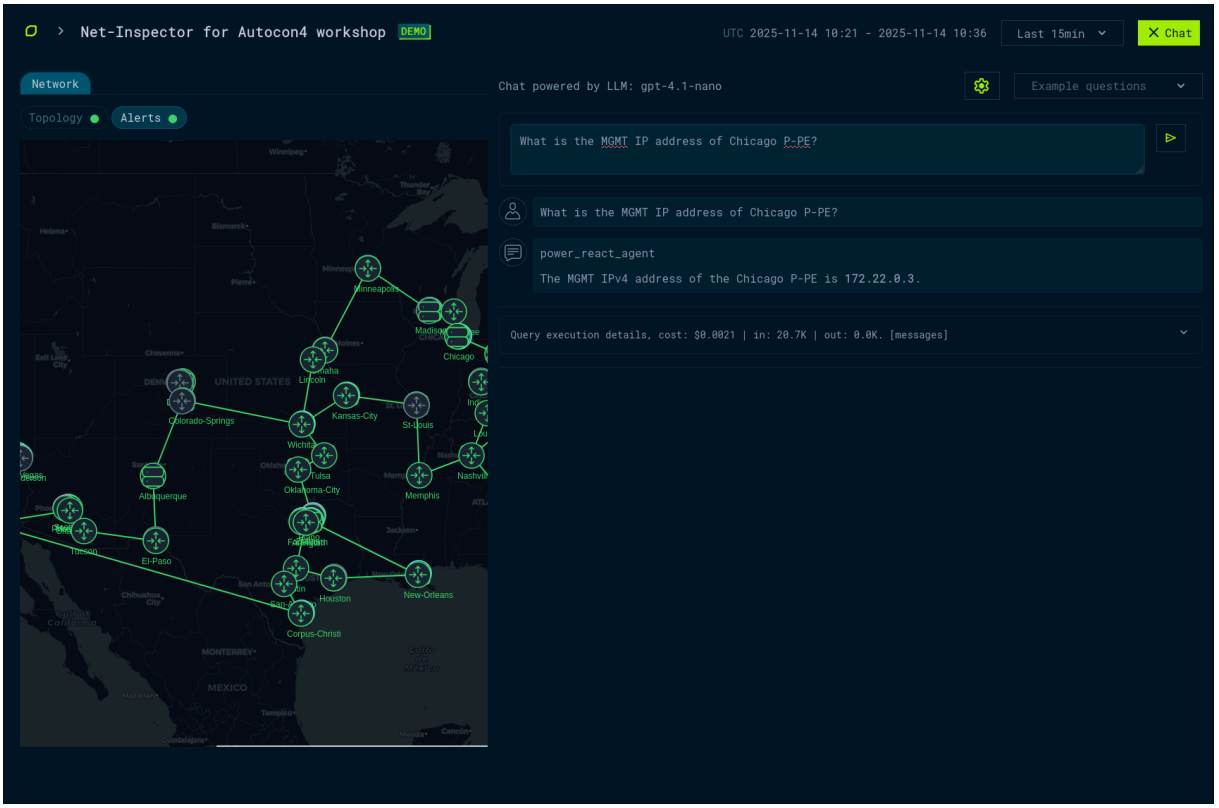
   `What is the MGMT IP address of Chicago P-PE?`

5. You should see a progress window (showing selected AI model, MCP agents etc):



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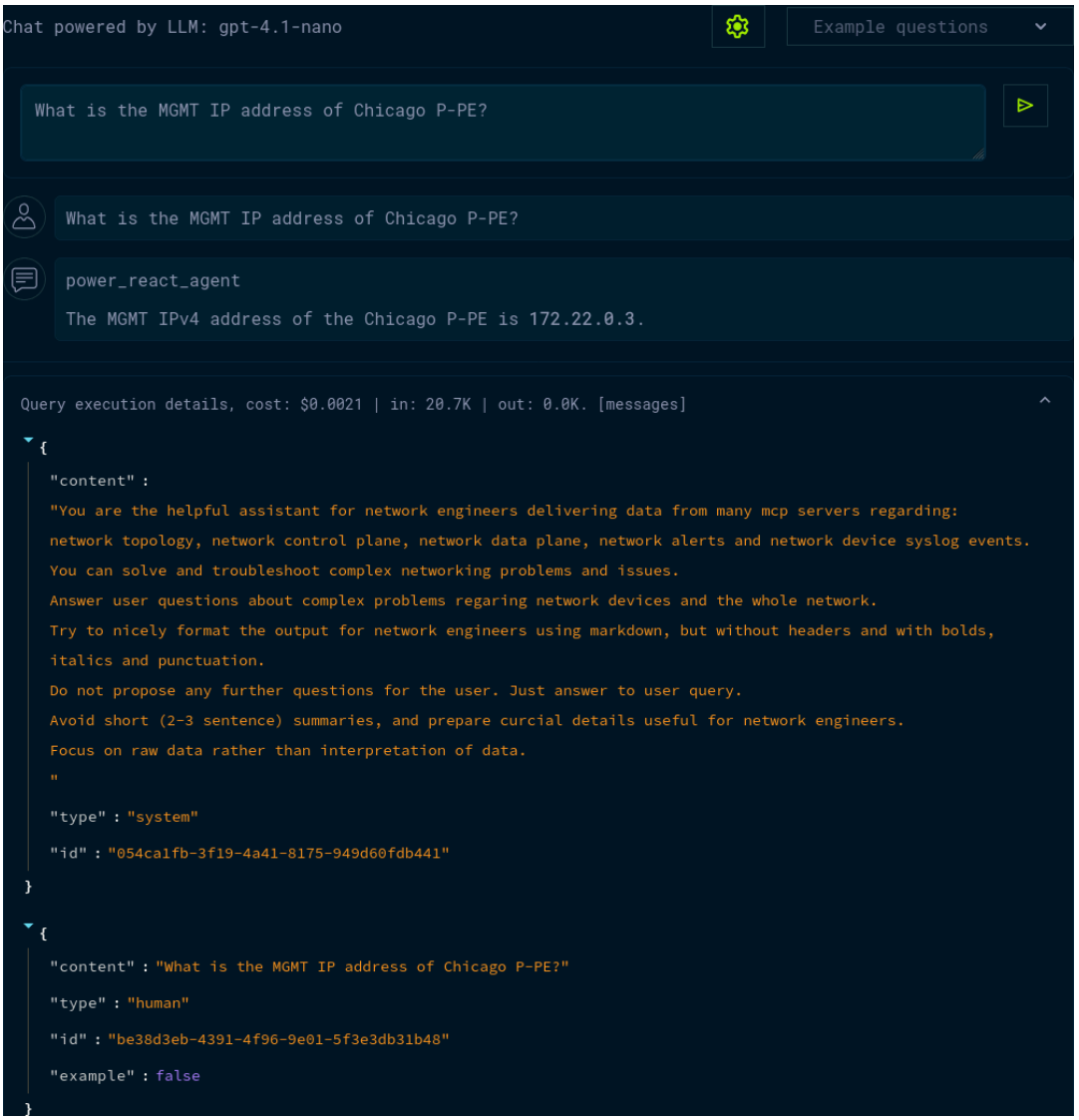
6. After all steps are finished, response is provided:



Note: Due to the nature of Streamlit do not use the Net-Inspector app while a query is processing.

7. Response can be expanded to show AI reasoning:

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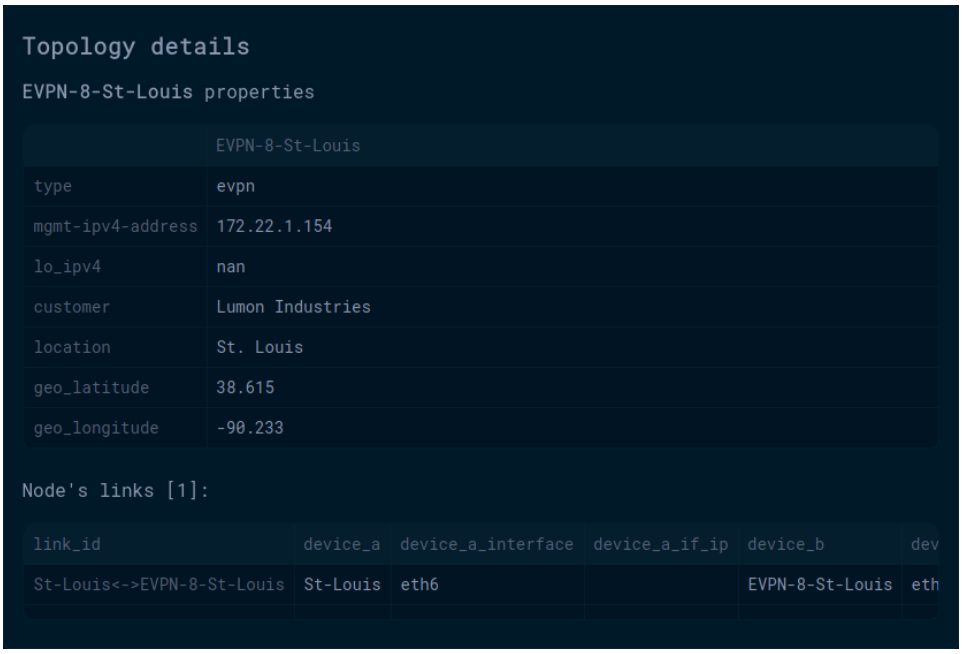


8. Later on Chat can be closed in order to get back to Topology and Alerts by using the “X Chat” button.

   Note: In order to view alerts (or topology details) you must close the chat first.

9. The map panel on the left can be zoomed in/out using a mouse scroll. When you select node, the right panel will present details on selected node:

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Topology details

EVPN-8-St-Louis properties

	EVPN-8-St-Louis
type	evpn
mgmt-ipv4-address	172.22.1.154
lo_ipv4	nan
customer	Lumon Industries
location	St. Louis
geo_latitude	38.615
geo_longitude	-90.233

Node's links [1]:

link_id	device_a	device_a_interface	device_a_if_ip	device_b	dev
St-Louis<->EVPN-8-St-Louis	St-Louis	eth6		EVPN-8-St-Louis	eth

10. Similarly selecting link will present details on selected link:



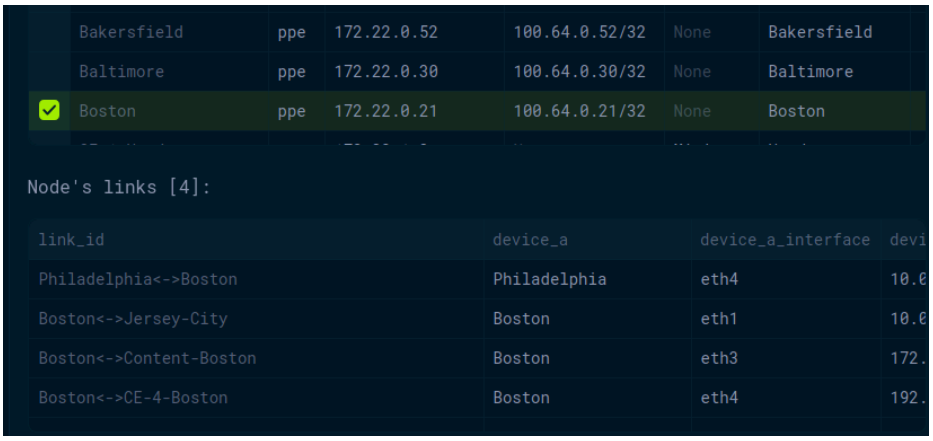
Topology details

Memphis<->St-Louis properties

	Memphis<->St-Louis
device_a	Memphis
device_a_interface	eth2
device_a_if_ip	10.0.0.245/30
device_a_mac	aa:c1:ab:f8:53:ea
device_b	St-Louis
device_b_interface	eth1
device_b_if_ip	10.0.0.246/30
device_b_mac	aa:c1:ab:68:43:ad

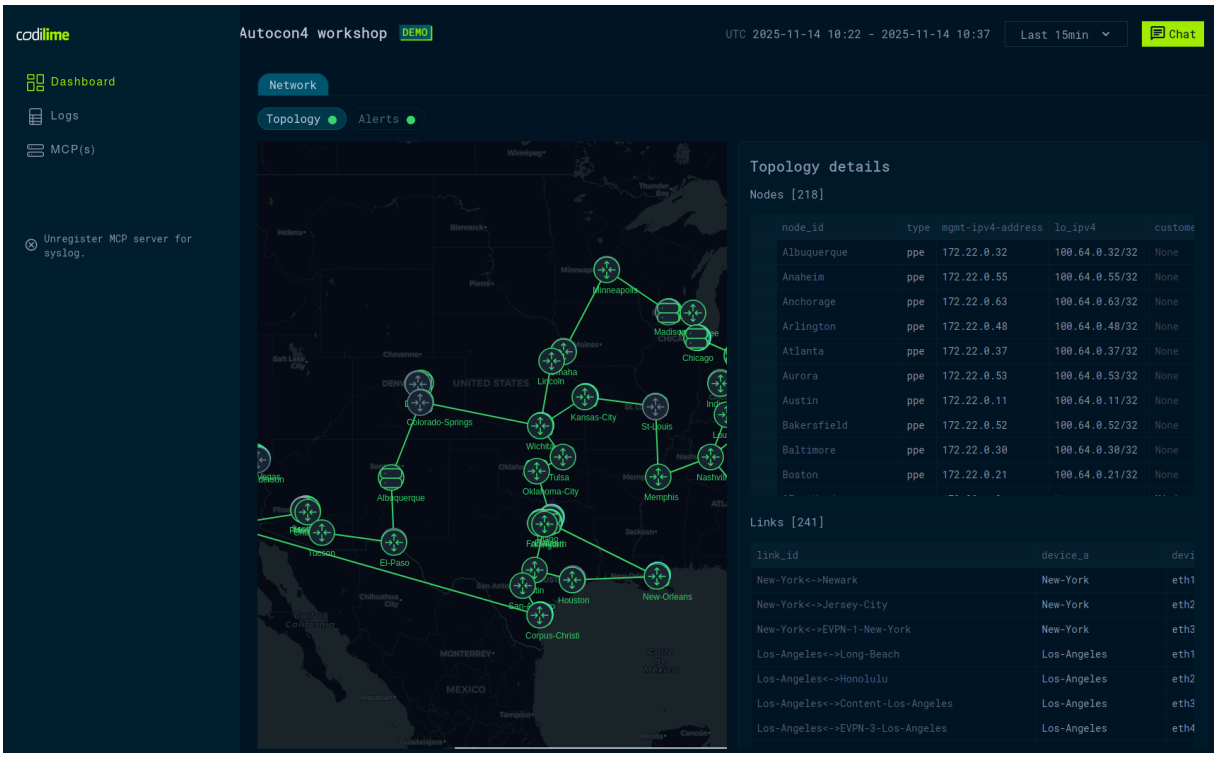
11. Alternatively when full nodes list is displayed you can click on the “invisible” (because of bug in Streamlit) checkbox left of the `node_id` to display links of that node:

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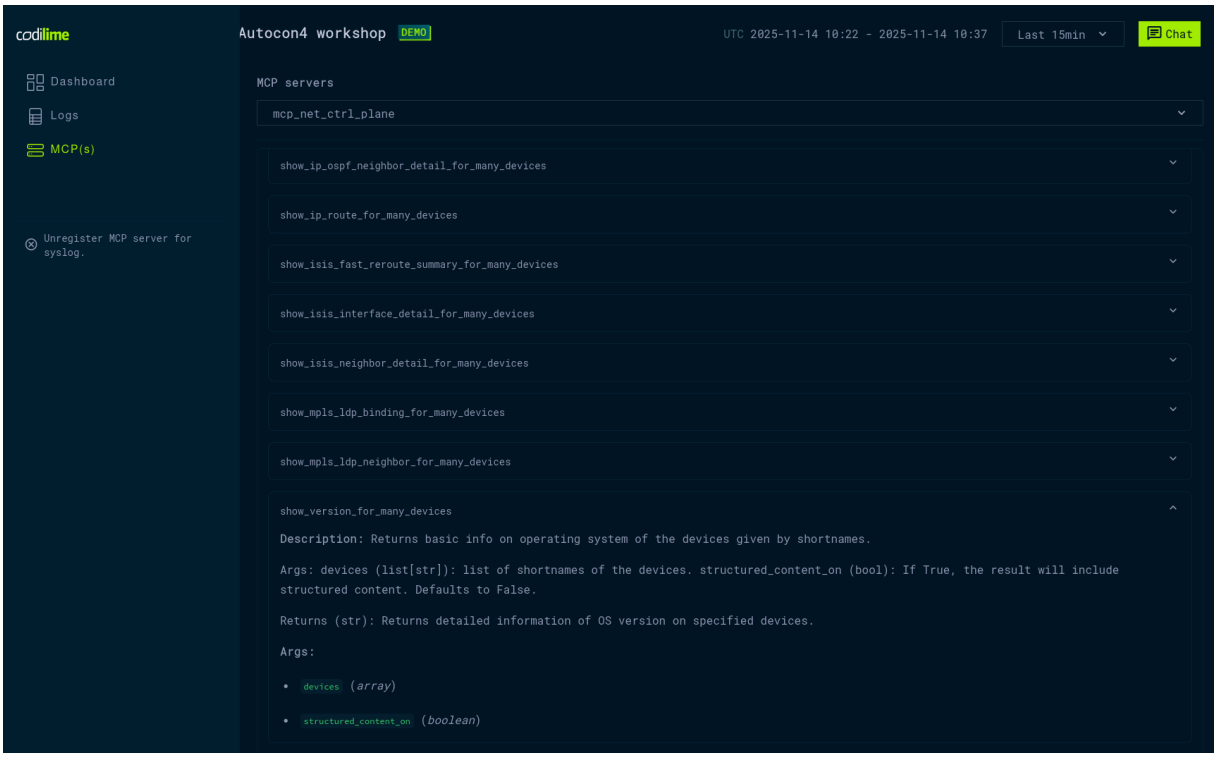
12. Cog wheel  over the chat allows users to change the AI model between `4.1-nano`, `4.1-mini`, `5-nano` and `5-mini`. Different models can provide better (or worse) responses.
13. Clicking the  symbol in top left corner, opens the menu with addition content:

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14. Logs contain events from the devices (syslog). At this moment no MCP server is registered. How to register the MCP log server will be discussed after the first exercise.
15. MCP shows existing MCP servers and tools that AI agents can use along with their description:

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[OPTIONAL] What the VM contains

The following chapter is mostly optional; however, you should familiarize yourself with the section about accessing Containerlab devices, as this knowledge will be needed later on.

Directory contents

Note: Although the user will have access to the scripts we highly recommend not to view them as this will make exercises less challenging.

Note: Please do not execute any scripts or change the contents of any file at this time.

In order to get familiar with the VM content, you can also use the `ls` command to list the directories that will be used during the workshop:

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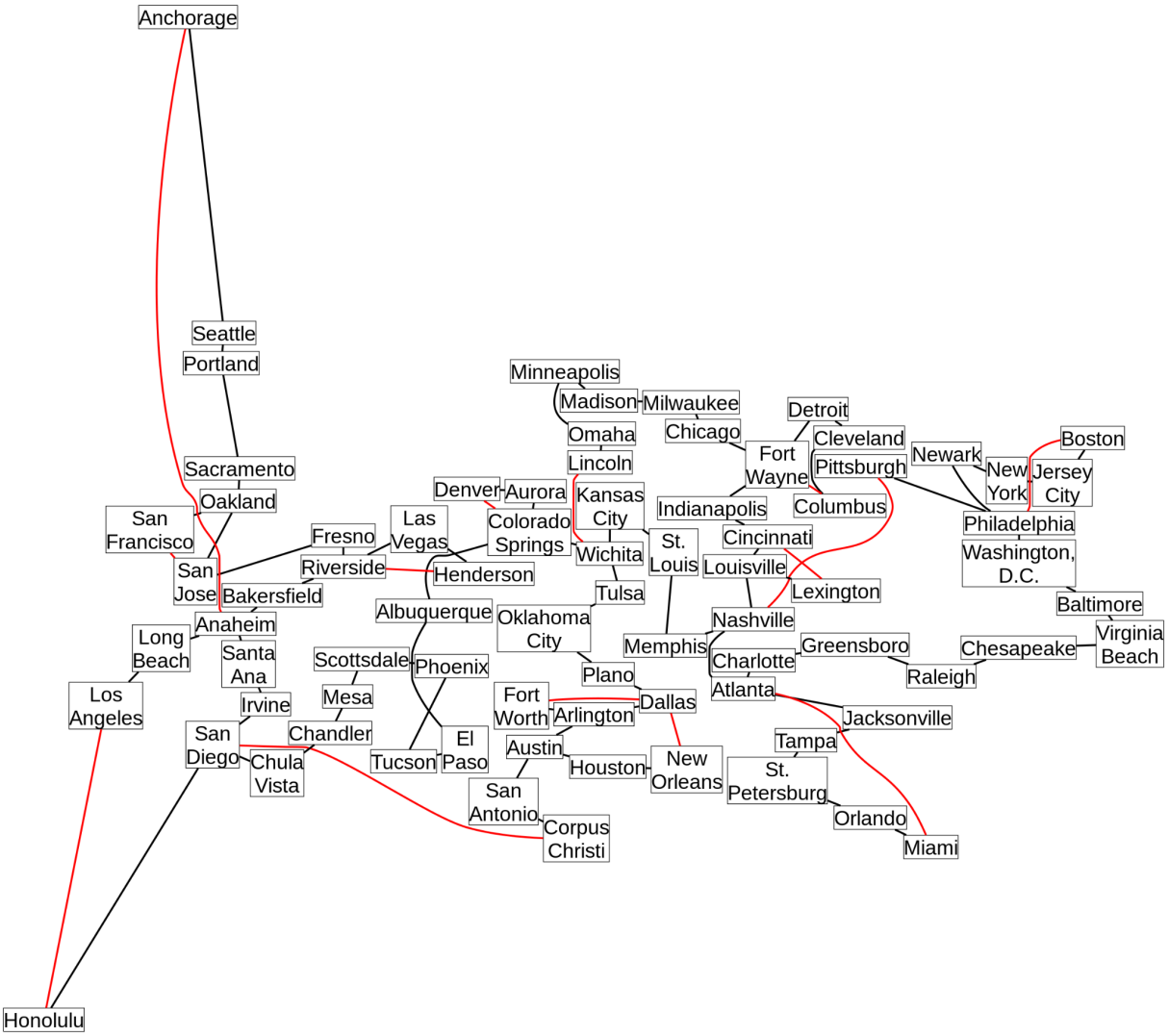
- **clab/** - directory with a network model used by containerlab
- **clab_backup/** - backup directory with a network model used by containerlab
- **clab/scripts/case*_fail.sh** - scripts for simulating failures in the network during the workshop
- **clab/scripts/case*_fix.sh** - scripts for network recovery after each hands-on part
- **clab/scripts/case*_*** - additional tools/files used for scenarios during the workshop. Should not be used directly.

Network topology overview

The diagram below presents the network topology to be used during the workshop. This is a rather large ISP topology with P/PE, CE routers and Content endpoints emulating customer endpoints and servers. There are several routing protocols running such as BGP, ISIS, LDP, BFD along with MPLS and VxLAN. The entire environment will come installed and ready to use on VMs running in a public cloud. To avoid any licensing issues and to save on resources, we decided to use open-source OS such as Alpine for Content nodes and FRR for PEs and CEs.

Note: Although full administrative access is granted to each device participating in the topology, we strongly recommend that you do not change existing configurations unless explicitly asked to do so.

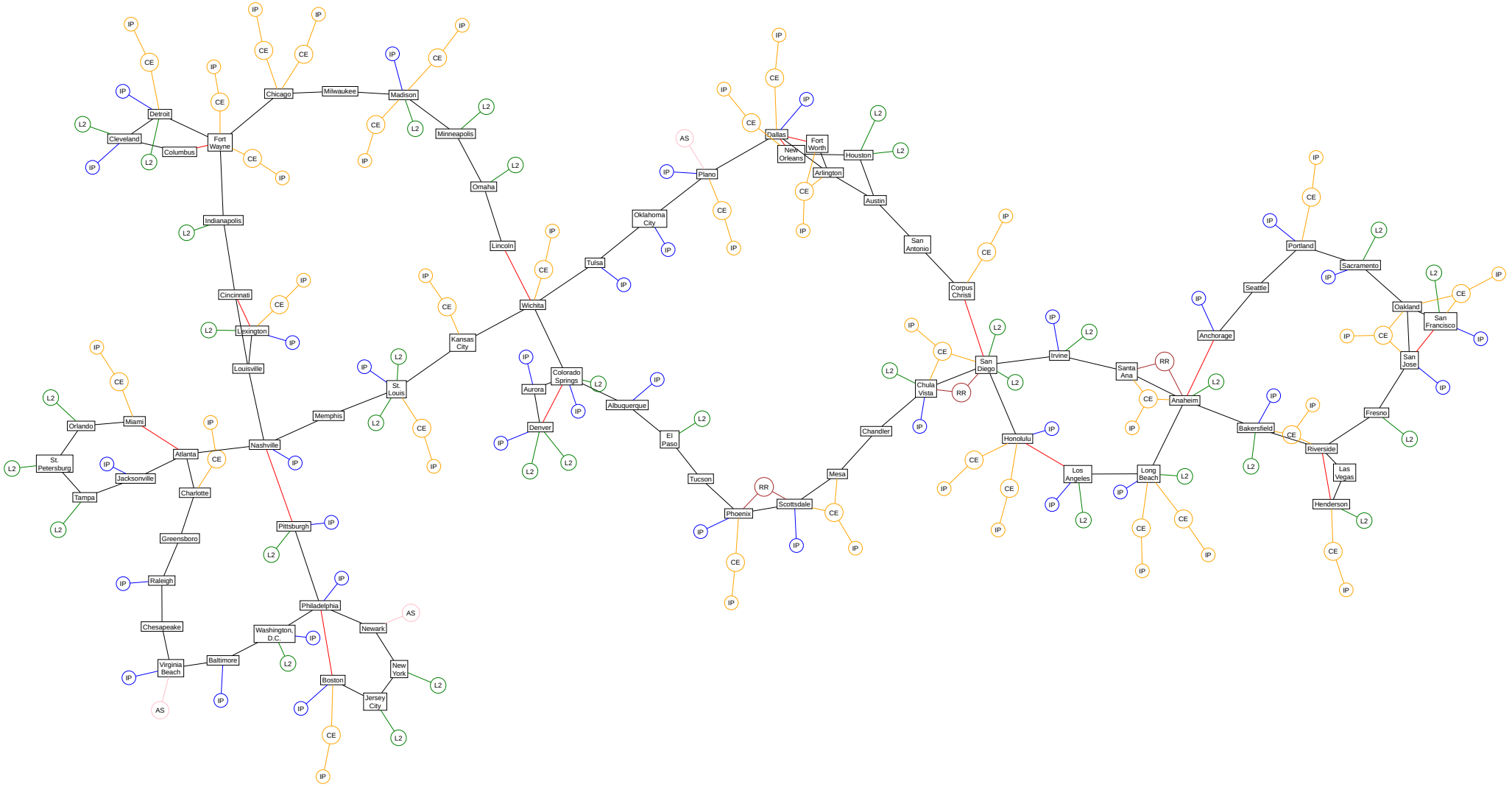
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The above topology shows cities and their connectivity between. Services such as Content/L3VN/EVPN are not included in order to keep drawing readable.

Diagram below shows a different approach to the topology, with geo-location removed and simplicity/readability in mind (IP endpoints are Content environments, CE are terminating L3VPN and L2 nodes are EVPN environments):

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Cleveland
Columbus
Fort Wayne
Chicago
Milwaukee
Madison
Minneapolis
Omaha
Lincoln
Wichita
Tulsa
Oklahoma City
Plano
Dallas
Fort Worth
Arlington
Houston
Austin
San Antonio
Corpus Christi
San Diego
Chula Vista
Irvine
Santa Ana
Anaheim
Long Beach
Los Angeles
Honolulu
Chandler
Mesa
Scottsdale
Phoenix
Tucson
El Paso
Albuquerque
Colorado Springs
Aurora
Denver
Kansas City
St. Louis
Memphis
Nashville
Louisville
Cincinnati
Lexington
Indianapolis
Charlotte
Atlanta
Miami
Orlando
St. Petersburg
Tampa
Jacksonville
Greensboro
Raleigh
Chesapeake
Virginia Beach
Washington, D.C.
Baltimore
Philadelphia
Pittsburgh
Newark
New York
Jersey City
Boston
Portland
Seattle
Anchorage
Sacramento
Oakland
San Francisco
San Jose
Fresno
Bakersfield
Riverside
Las Vegas
Henderson
IP
CE
L2
RR
AS

IP addresses and network used in topology

For simplicity and ease of access to following network blocks were used during topology construction:

172.22.0.0/22 - management IPs for all devices in the network. This prefix is directly reachable from the lab-X VM.

100.64.0.0/24 - used for loopbacks for P/PE devices in the network

100.65.0.0/24 - used for loopbacks for BGP RR devices in the network

10.0.0.0/16 (divided into /30 subnets) - used for P2P connections between P/PE devices

192.168.0.0/16 (divided into /30 subnets) - used for P2P connections between P/PE devices and CE devices (L3VPN customers)

172.24.0.0/13 (divided into /24 subnets) - used for addressing Content endpoints (emulating client/servers) in selected cities. One prefix per **location** is used.

172.17.0.0/16 (divided into /26 subnets) - used for addressing L3VPN customer clients/servers behind CE in selected cities. One prefix per **location** is used.

172.16.0.0/16 (divided into /25 subnets) - used for addressing EVPN customer clients/servers in selected cities. One prefix per **customer** is used.

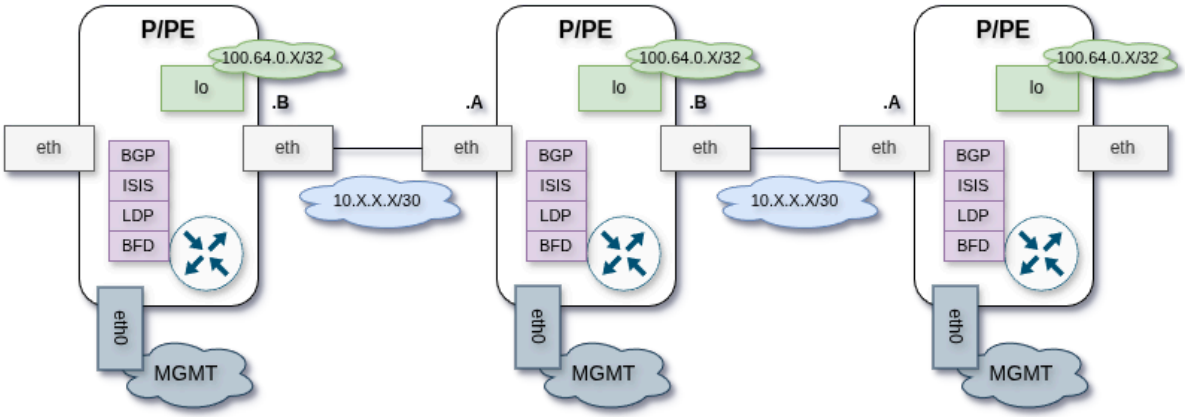
All prefixes from 172.16.0.0/12 pool are selected randomly.

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Access to network devices via SSH

For all devices login and password is **root** / **root.** Access is provided via SSH from the VM. You can use either their hostnames or IP addresses. Management IP addresses of each device are listed at the end of each following chapter. Below one can find types of devices that are used in workshop topology.

P/PE devices:



P/PE devices are running FRR with the following protocols enabled:

- BGP
- ISIS
- LDP
- BFD

Their function is to provide services and forward traffic between them.

Each P/PE device is assigned to a dedicated city and named after that city.

Each P/PE device can have zero or more functionalities (customers/devices etc) attached to it such as (for more detailed description see next chapters):

- **RR** - those are BGP Route Reflector devices

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- **eBGP** - those are devices emulating external peering
- **Content** - those are environments emulating home customers and content providers (in real ISP environment those would Internet users and Internet servers)
- **EVPN** - those are environments divided by groups (each group representing one customer) connected on Layer2 using EVPN
- **CE + CE-IP** - those are environments divided by groups (each group representing one customer) connected on Layer3 using L3VPN (based on MPLS)

Note: Device logs are located in `/var/log/messages` (accessible from shell)

Logging into device

One can login into each P/PE device either by DNS name (`ssh root@atlanta`) or by IP (`ssh root@172.22.0.37`). Login and password is **root / root.** In order to launch router CLI one must execute the **`vtysh`** command after login.

Below one can find list of all P/PE devices and their management IPs in alphabetical order:

```
Albuquerque          172.22.0.32
Anaheim              172.22.0.55
Anchorage            172.22.0.63
Arlington            172.22.0.48
Atlanta              172.22.0.37
Aurora               172.22.0.53
Austin               172.22.0.11
Bakersfield          172.22.0.52
Baltimore            172.22.0.30
Boston               172.22.0.21
Chandler             172.22.0.75
Charlotte            172.22.0.16
```

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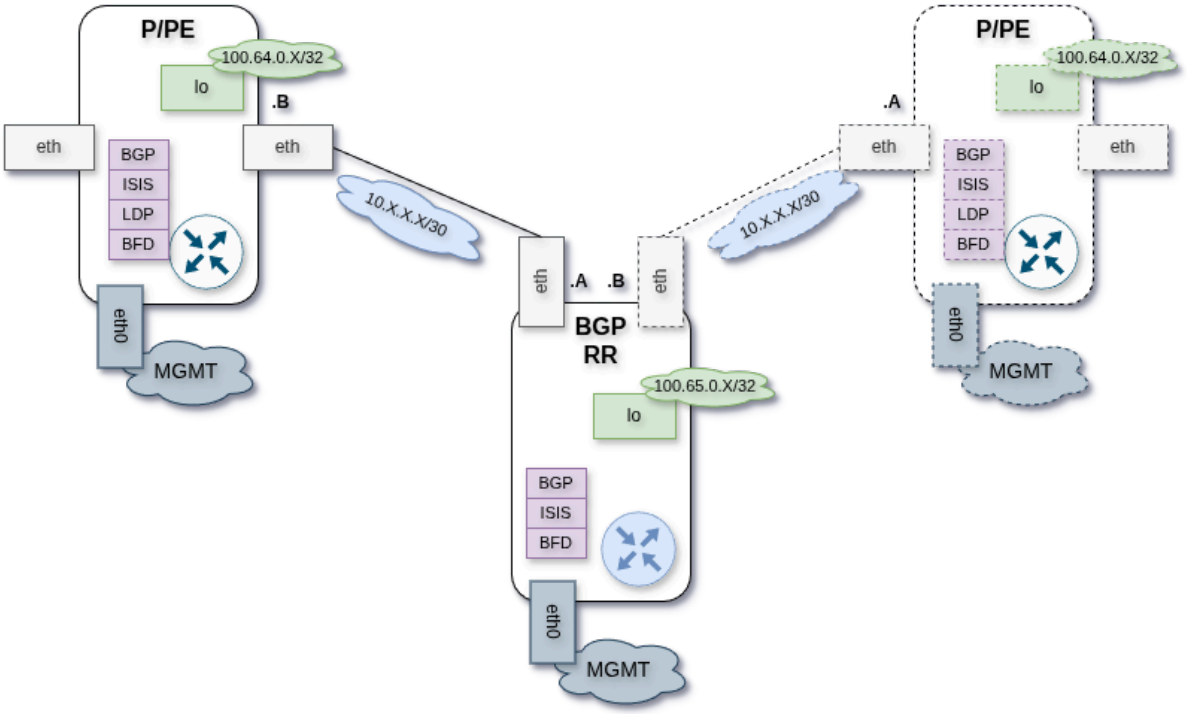
```
Chesapeake          172.22.0.79
Chicago             172.22.0.3
Chula-Vista         172.22.0.74
Cincinnati          172.22.0.64
Cleveland           172.22.0.54
Colorado-Springs    172.22.0.39
Columbus            172.22.0.14
Corpus-Christi      172.22.0.59
Dallas              172.22.0.9
Denver              172.22.0.19
Detroit             172.22.0.23
El-Paso             172.22.0.22
Fort-Wayne          172.22.0.77
Fort-Worth          172.22.0.13
Fresno              172.22.0.34
Greensboro          172.22.0.70
Henderson           172.22.0.65
Honolulu            172.22.0.56
Houston             172.22.0.4
Indianapolis        172.22.0.15
Irvine              172.22.0.69
Jacksonville        172.22.0.12
Jersey-City         172.22.0.73
Kansas-City         172.22.0.38
Las-Vegas           172.22.0.26
Lexington           172.22.0.60
Lincoln             172.22.0.71
Long-Beach          172.22.0.44
Los-Angeles         172.22.0.2
Louisville          172.22.0.28
Madison             172.22.0.78
Memphis             172.22.0.29
Mesa                172.22.0.35
Miami               172.22.0.40
```

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```
Milwaukee           172.22.0.31
Minneapolis         172.22.0.46
Nashville           172.22.0.24
Newark              172.22.0.68
New-Orleans         172.22.0.51
New-York            172.22.0.1
Oakland             172.22.0.45
Oklahoma-City       172.22.0.27
Omaha               172.22.0.41
Orlando             172.22.0.66
Philadelphia        172.22.0.6
Phoenix             172.22.0.5
Pittsburgh          172.22.0.62
Plano               172.22.0.67
Portland            172.22.0.25
Raleigh             172.22.0.42
Riverside           172.22.0.57
Sacramento          172.22.0.36
San-Antonio         172.22.0.7
San-Diego           172.22.0.8
San-Francisco       172.22.0.17
San-Jose            172.22.0.10
Santa-Ana           172.22.0.58
Scottsdale          172.22.0.76
Seattle             172.22.0.18
St-Louis            172.22.0.61
St-Petersburg       172.22.0.72
Tampa               172.22.0.49
Tucson              172.22.0.33
Tulsa               172.22.0.47
Virginia-Beach      172.22.0.43
Washington-DC       172.22.0.20
Wichita             172.22.0.50
```

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BGP RR devices



BGP RR devices are running FRR with the following protocols enabled:

- BGP
- ISIS
- BFD

Their function is to act as a BGP Route Reflector for the whole network. Using ISIS high metric and having MPLS/LDP disabled there are excluded from forwarding the traffic.

Each BGP RR device is assigned to a dedicated city and named after that city. If there is another city close by, BGP RR is also connected to it.

Note: Device logs are located in `/var/log/messages` (accessible from shell)

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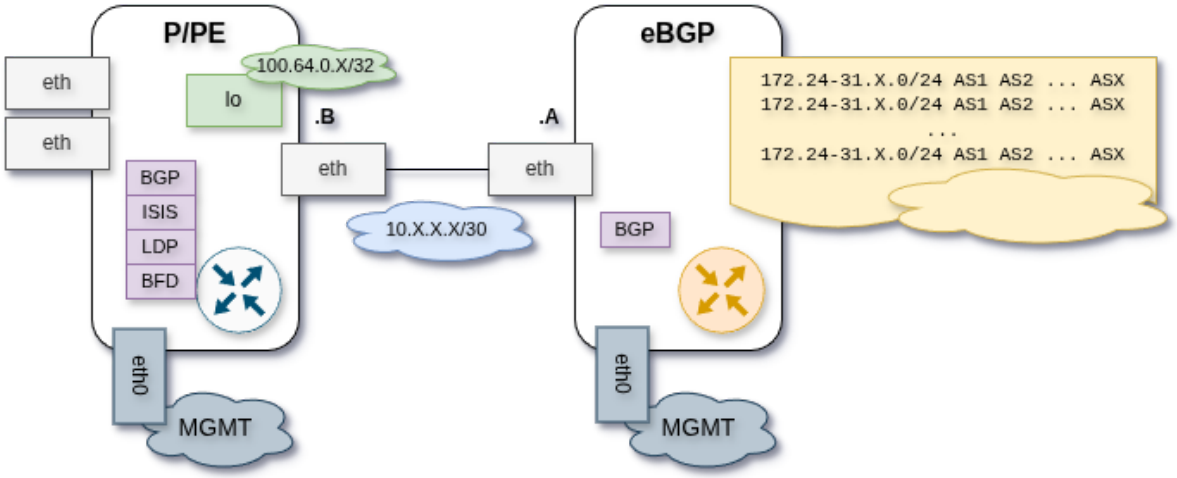
Logging into device

One can login into each BGP-RR device either by DNS name (`ssh root@rr-anaheim`) or by IP (`ssh root@172.22.2.131`). Login and password is **root / root.** In order to launch router CLI one must execute the **`vtysh`** command after login.

Below one can find list of all BGP RR devices and their management IPs in alphabetical order:

```
RR-Anaheim                    172.22.2.131
RR-Chula-Vista                172.22.2.129
RR-Phoenix                    172.22.2.130
```

eBGP devices



eBGP devices are running Bird 3.x with the following protocols enabled:

- BGP

Their function is to act as an external BGP peer and saturate the network with multiple prefixes (with different AS paths).

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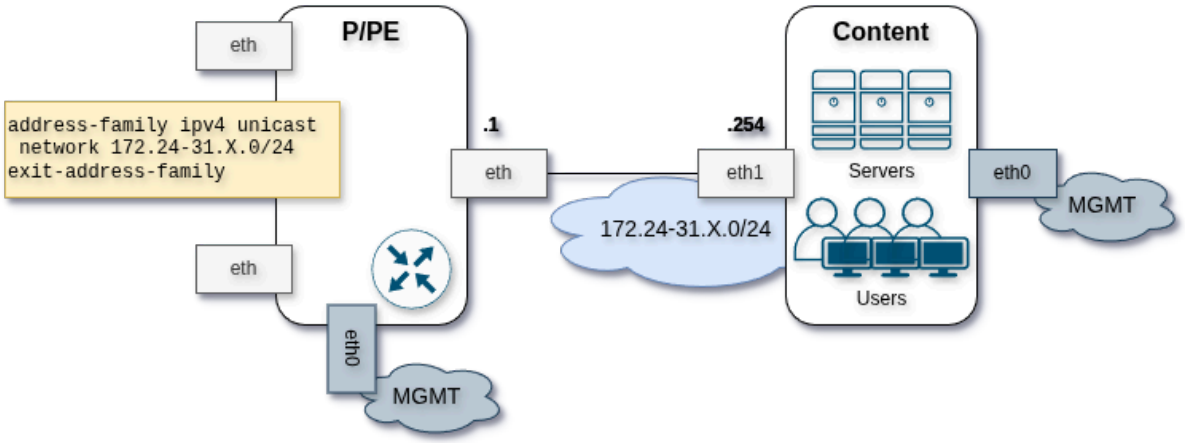
Each eBGP device is assigned to a dedicated city and named after that city.

Below one can find list of all eBGP devices and their management IPs in alphabetical order:

```
eBGP-Newark                 172.22.2.194
eBGP-Plano                  172.22.2.195
eBGP-Virginia-Beach         172.22.2.193
```

Note: Although one can login into eBGP devices, those devices won't be used in the workshop and should be treated as external devices.

Content sites



Content containers are emulating environments with customers and content providers in selected cities. They are attached to a PE router with a direct link that has a /24 network on it. That network is directly advertised via BGP to the rest of the network.

For the scalability reasons, the internal config is kept as simple as possible. Each environment has multiple random IPs (there is always one fixed IP ending with .254) associated with eth1 interface:

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```
Content-Anchorage:/# ip -br a
lo               UNKNOWN        127.0.0.1/8 ::1/128
eth0@if6847      UP             172.22.2.6/22
vrf_mng          UP
eth1@if6866          UP                      172.28.14.254/24  172.28.14.129/24
172.28.14.79/24     172.28.14.211/24     172.28.14.157/24     172.28.14.170/24
fe80::a8c1:abff:feb6:d008/64
```

For each IP there is web server (nginx) listening and serving two sites:

```
Content-Anchorage:/# curl 172.28.14.129
<!DOCTYPE html>
<html>
<head>
<title> Content-Anchorage /title>
</head>
<body>

<h1>This is a NGINX at:</h1>
<p>Content-Anchorage</p>

</body>
</html>

Content-Anchorage:/# curl 172.28.14.129/big.html
<!DOCTYPE html>
<html>
<head>
<title> Content-Anchorage /title>
</head>
<body>

<p>Lorem ipsum dolor sit amet,
[...]
```

Each content site is also hosting iperf3 server and client. They are used for background traffic generation between sites.

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Note: Device logs (WWW server) are located in `/var/log/nginx/` (accessible from shell). They are also sent to AI for analysis.

Logging into device

One can login into each Content environment either by DNS name (`ssh root@content-aurora`) or by IP (`ssh root@172.22.2.25`). Login and password is **root / root.**

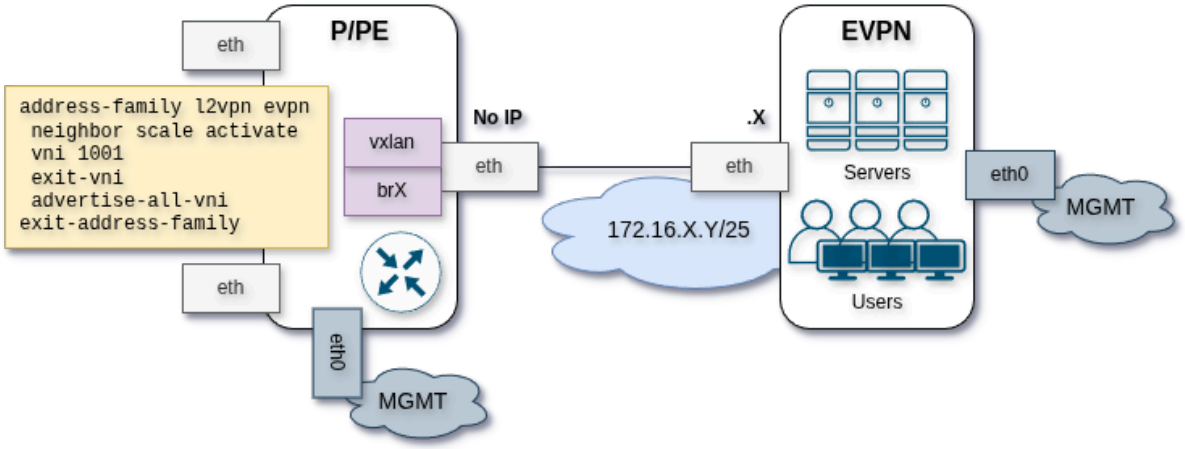
Below one can find list of all Content environment and their management IPs in alphabetical order:

```
Content-Albuquerque          172.22.2.14
Content-Anchorage            172.22.2.6
Content-Aurora               172.22.2.25
Content-Bakersfield          172.22.2.35
Content-Baltimore            172.22.2.28
Content-Boston               172.22.2.8
Content-Chula-Vista          172.22.2.2
Content-Cleveland            172.22.2.30
Content-Colorado-Springs     172.22.2.20
Content-Dallas               172.22.2.19
Content-Denver               172.22.2.24
Content-Detroit              172.22.2.21
Content-Honolulu             172.22.2.18
Content-Irvine               172.22.2.23
Content-Jacksonville         172.22.2.29
Content-Lexington            172.22.2.5
Content-Long-Beach           172.22.2.26
Content-Los-Angeles          172.22.2.3
Content-Madison              172.22.2.11
Content-Nashville            172.22.2.33
Content-Oklahoma-City        172.22.2.4
Content-Philadelphia         172.22.2.16
Content-Phoenix              172.22.2.9
Content-Pittsburgh           172.22.2.1
Content-Plano                172.22.2.10
Content-Portland             172.22.2.22
Content-Raleigh              172.22.2.12
Content-Sacramento           172.22.2.7
Content-San-Francisco        172.22.2.27
```

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```
Content-San-Jose                172.22.2.13
Content-Scottsdale              172.22.2.17
Content-St-Louis                172.22.2.34
Content-Tulsa                   172.22.2.15
Content-Virginia-Beach          172.22.2.32
Content-Washington-DC           172.22.2.31
```

EVPN sites



EVPN containers are emulating customer sites connected to each other using Layer 2 via EVPN. They are attached to a PE router with a direct link that has a /25 network on the customer side and no IP on the PE side. On PE that link is attached to the bridge which is a part of EVPN.

Each container has a random IP on eth1 interface:

```
EVPN-10-Omaha:/# ip -br a
lo               UNKNOWN        127.0.0.1/8 ::1/128
eth0@if7208      UP             172.22.1.160/22
vrf_mng          UP
eth1@if6998      UP             172.16.4.187/25 fe80::a8c1:abff:fece:9e2d/64
```

For each IP there is a web server (nginx) iperf3 server listening and serving requests.

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Note: Device logs (WWW server) are located in `/var/log/nginx/` (accessible from shell). They are also sent to AI for analysis.

Logging into device

One can login into each EVPN environment either by DNS name (`ssh root@EVPN-10-Omaha`) or by IP (`ssh root@172.22.1.160`). Login and password is **root / root.**

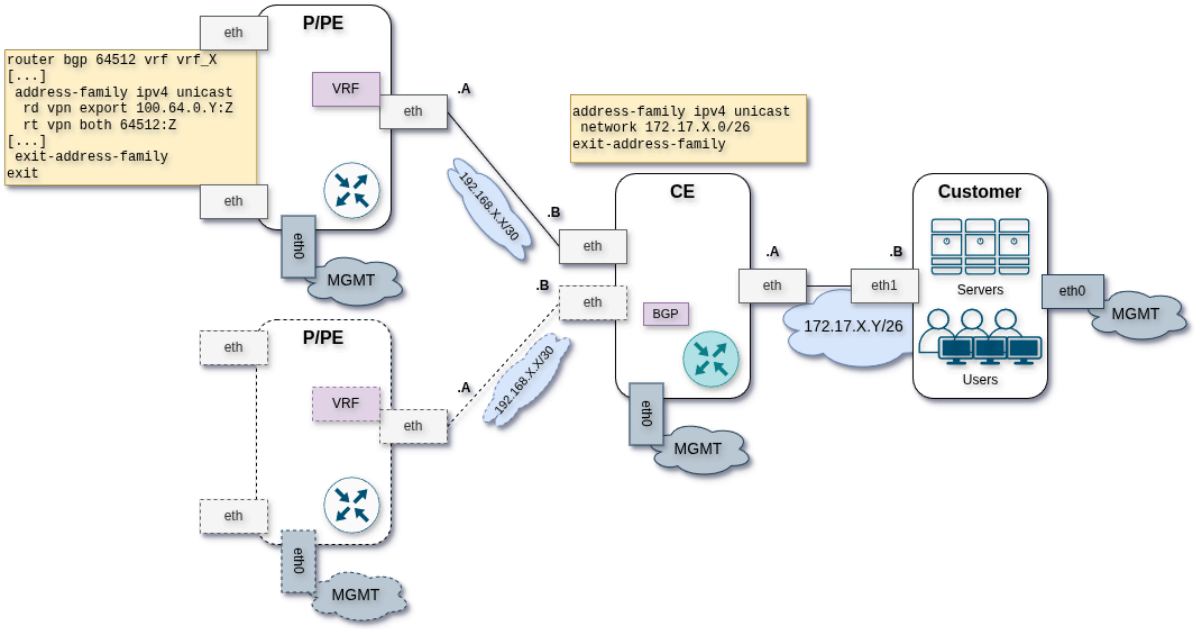
Below one can find a list of all EVPN environments and their management IPs in alphabetical order. Digit within site name points to a customer number:

```
EVPN-10-Omaha                172.22.1.160
EVPN-10-San-Diego            172.22.1.162
EVPN-10-San-Francisco        172.22.1.161
EVPN-1-Chula-Vista           172.22.1.130
EVPN-1-Fresno                172.22.1.131
EVPN-1-New-York              172.22.1.129
EVPN-2-Anaheim               172.22.1.132
EVPN-2-Denver                172.22.1.134
EVPN-2-Orlando               172.22.1.133
EVPN-2-San-Diego             172.22.1.135
EVPN-3-Irvine                172.22.1.136
EVPN-3-Los-Angeles           172.22.1.137
EVPN-4-Cleveland             172.22.1.138
EVPN-4-Houston               172.22.1.140
EVPN-4-Washington-DC         172.22.1.139
EVPN-5-Colorado-Springs      172.22.1.143
EVPN-5-Indianapolis          172.22.1.141
EVPN-5-Lexington             172.22.1.144
EVPN-5-Pittsburgh            172.22.1.142
EVPN-6-Bakersfield           172.22.1.147
EVPN-6-Denver                172.22.1.145
EVPN-6-Houston               172.22.1.146
EVPN-6-Madison               172.22.1.148
EVPN-7-Detroit               172.22.1.149
EVPN-7-Henderson             172.22.1.151
EVPN-7-Long-Beach            172.22.1.150
EVPN-7-St-Louis              172.22.1.152
EVPN-8-El-Paso               172.22.1.153
EVPN-8-Sacramento            172.22.1.155
```

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```
EVPN-8-St-Louis                172.22.1.154
EVPN-9-Jersey-City             172.22.1.157
EVPN-9-Minneapolis             172.22.1.156
EVPN-9-St-Petersburg           172.22.1.158
EVPN-9-Tampa                   172.22.1.159
```

L3VPN sites



L3VPN sites are built from 2 parts:

- CE device (eg. `CE-4-boston`)
- Customer servers/clients (eg. `CE-IP-4-boston`)

A CE device can be connected to one or two cities and is named after the first city (just like in BGP-RR naming). Between CE and P/PE there is an eBGP session advertising a single customer network into the L3VPN network. CE device is running FRR with a simple configuration.

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The customer container is emulating the customer site. Those sites are connected to each other (via CE) using Layer3 within a single customer. There is no VRF leaking between the Internet and L3VPN sites configured. Customer containers are (in terms of configuration and connectivity) very similar to Content containers.

Each customer container has a random IPs on eth1 interface:

```
ce-ip-7-madison:/# ip -br a
lo               UNKNOWN        127.0.0.1/8 ::1/128
eth0@if39066     UP             172.22.2.58/22
vrf_mng          UP
eth1@if39079         UP                    172.17.176.254/26  172.17.176.251/26
172.17.176.219/26   172.17.176.231/26   172.17.176.207/26   172.17.176.212/26
172.17.176.216/26 172.17.176.
198/26 172.17.176.206/26 172.17.176.233/26 fe80::a8c1:abff:fe1e:72/64
```

For each IP there is a web server (nginx) iperf3 server listening and serving requests.

Note: On customer sites logs of WWW server) are located in `/var/log/nginx/` (accessible from shell). On CE devices there is `/var/log/messages` file. All events are sent to AI for analysis.

Logging into device

One can login into each CE and Customer environment either by DNS name (`ssh root@ce-ip-4-boston`) or by IP (`ssh root@172.22.2.45`). Login and password is **root / root.** In order to launch router CLI on CE devices one must execute the **`vtysh`** command after login.

Below one can find a list of all CE and customer environments and their management IPs in alphabetical order. Digit within name points to a customer number:

```
CE-10-Bakersfield              172.22.1.31
CE-10-Corpus-Christi           172.22.1.32
```

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```
CE-10-Phoenix                172.22.1.30
CE-1-Henderson               172.22.1.3
CE-1-Kansas-City             172.22.1.2
CE-1-Lexington               172.22.1.1
CE-2-Chicago                 172.22.1.6
CE-2-Fort-Wayne              172.22.1.4
CE-2-Miami                   172.22.1.5
CE-2-Scottsdale              172.22.1.7
CE-3-Chula-Vista             172.22.1.9
CE-3-Long-Beach              172.22.1.8
CE-4-Boston                  172.22.1.10
CE-4-Charlotte               172.22.1.11
CE-4-Madison                 172.22.1.12
CE-4-Oakland                 172.22.1.13
CE-5-Fort-Worth              172.22.1.15
CE-5-Honolulu                172.22.1.16
CE-5-Portland                172.22.1.14
CE-6-Anaheim                 172.22.1.19
CE-6-Fort-Wayne              172.22.1.18
CE-6-New-Orleans             172.22.1.17
CE-6-San-Jose                172.22.1.20
CE-7-Madison                 172.22.1.23
CE-7-Plano                   172.22.1.21
CE-7-St-Louis                172.22.1.22
CE-8-Chicago                 172.22.1.24
CE-8-Detroit                 172.22.1.27
CE-8-Honolulu                172.22.1.25
CE-8-Long-Beach              172.22.1.26
CE-9-Dallas                  172.22.1.29
CE-9-Wichita                 172.22.1.28

CE-IP-10-Bakersfield         172.22.2.66
CE-IP-10-Corpus-Christi      172.22.2.67
CE-IP-10-Phoenix             172.22.2.65
CE-IP-1-Henderson            172.22.2.38
CE-IP-1-Kansas-City          172.22.2.37
CE-IP-1-Lexington            172.22.2.36
CE-IP-2-Chicago              172.22.2.41
CE-IP-2-Fort-Wayne           172.22.2.39
CE-IP-2-Miami                172.22.2.40
CE-IP-2-Scottsdale           172.22.2.42
CE-IP-3-Chula-Vista          172.22.2.44
CE-IP-3-Long-Beach           172.22.2.43
CE-IP-4-Boston               172.22.2.45
CE-IP-4-Charlotte            172.22.2.46
CE-IP-4-Madison              172.22.2.47
```

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```
CE-IP-4-Oakland              172.22.2.48
CE-IP-5-Fort-Worth           172.22.2.50
CE-IP-5-Honolulu             172.22.2.51
CE-IP-5-Portland             172.22.2.49
CE-IP-6-Anaheim              172.22.2.54
CE-IP-6-Fort-Wayne           172.22.2.53
CE-IP-6-New-Orleans          172.22.2.52
CE-IP-6-San-Jose             172.22.2.55
CE-IP-7-Madison              172.22.2.58
CE-IP-7-Plano                172.22.2.56
CE-IP-7-St-Louis             172.22.2.57
CE-IP-8-Chicago              172.22.2.59
CE-IP-8-Detroit              172.22.2.62
CE-IP-8-Honolulu             172.22.2.60
CE-IP-8-Long-Beach           172.22.2.61
CE-IP-9-Dallas               172.22.2.64
CE-IP-9-Wichita              172.22.2.63
```

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Hands-on

Introduction - Obtaining information about the network

Objectives

- Get familiar with the UI and underlying network topology

Use AI to obtain network inventory Steps

Query the AI Agent

- Ask questions at the bottom of this chapter
- Observe the results and verify them against the network topology described in this guide
- If the output is not satisfactory, try repeating the same query or rephrasing it

Example queries:

```
Show ppe devices and their topological details

Show links and their topological details

How are CE devices connected in Boston?

Check connectivity from CE devices to New York, present
the results in tabular form

Get basic data on interfaces on PPE devices, present the
results in tabular form

Is there a direct connection between Anchorage and
Seattle? If so, present its details.

Where are BGP Route Reflectors located?
```

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Part 1a - Topology investigation (without syslog MCP)

Objectives

- Debug the lack of IGP/LDP redundancy issue (see detailed description)

Steps

Introduce a failure

Execute the following script in the VM shell:

```
$ ~/clab/scripts/case1_fail.sh
```

Investigate

This script will change the configuration between two (directly connected devices in the network). This change will:

1. keep the IP address intact on that link
2. keep the link state intact
3. remove all ISIS and LDP configuration from that link only **on one of the P/PE**

As a result, part of the network will lose its backup path for transit traffic (as there will be no ISIS / LDP protocols running over that link). Task of this case is to find which nodes has only one fully functional path

Restore the network to a normal state

Run the following script:

```
$ ~/clab/scripts/case1_fix.sh
```

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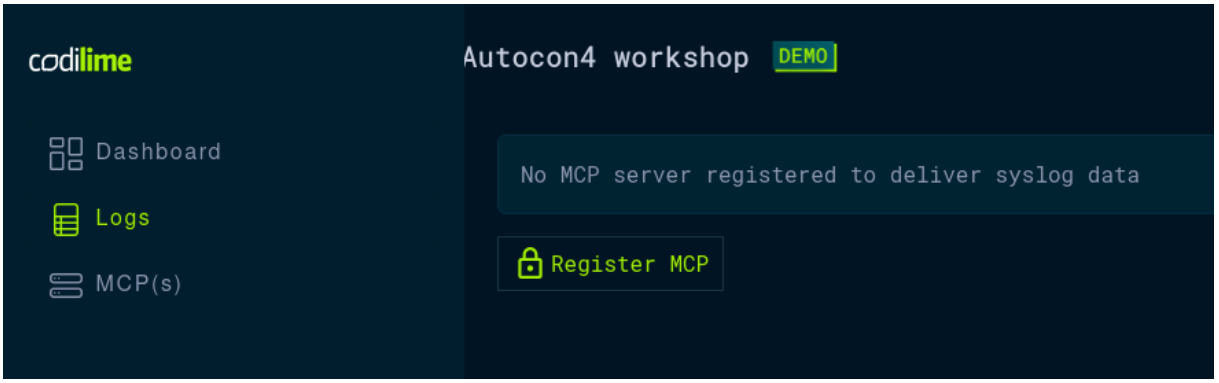
Part 1b - Topology investigation (with syslog MCP)

Objectives

- Enable logging MCP in UI
- Repeat debugging session but also using syslog
- This time concentrate not on topology status but on recent events regarding ISIS / LDP

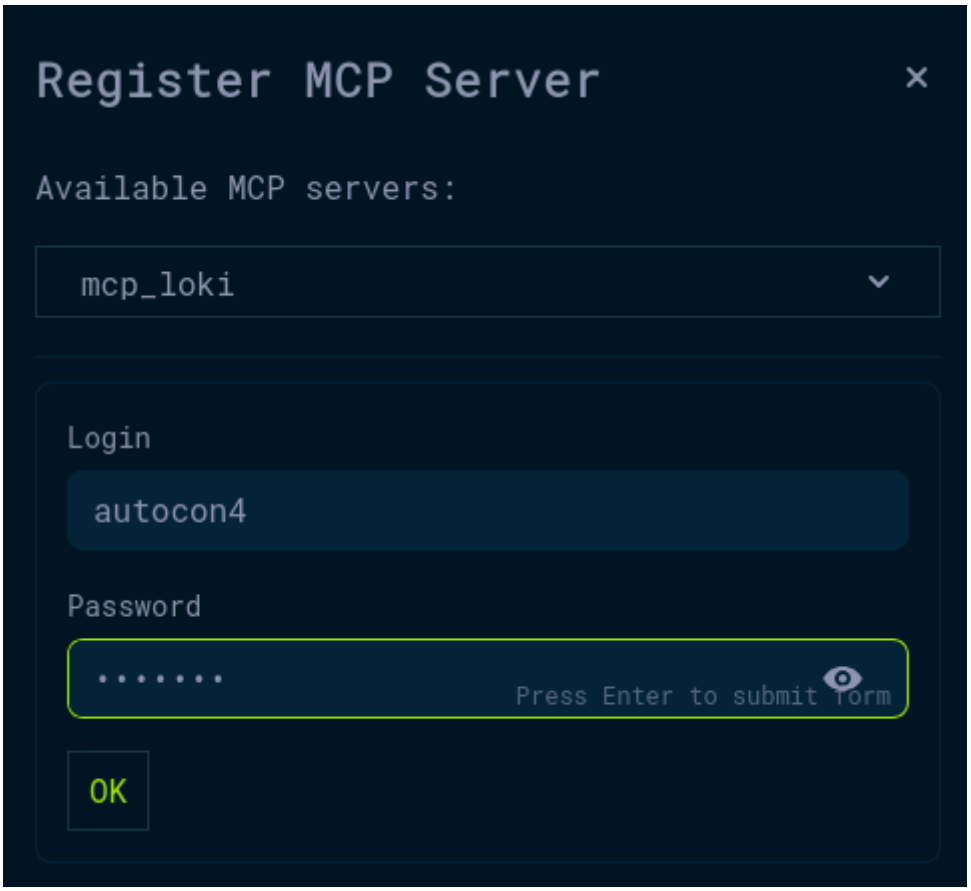
Enabling MCP logging

1. Open side menu using “>”:
2. Click on the Logs
3. Click on “Register MCP”



4. Enter autocon4 / autocon4 as Login and Password:

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5. Click OK to enable logging MCP

Workshop steps

Steps and issues are identical to **Part 1a**, however this time AI reasoning will be enhanced with syslog events from the devices, so questions about last events can be asked as well.

Note: After all steps remember to run the following script to restore the network to a normal state:

```
$ ~/clab/scripts/case1_fix.sh
```

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Part 2 - Traffic volume drop

Objectives

- Investigate sudden drop in traffic between two cities

Steps

Introduce a failure

Execute the following script in the VM shell:

```
$ ~/clab/scripts/case2_fail.sh
```

Investigate

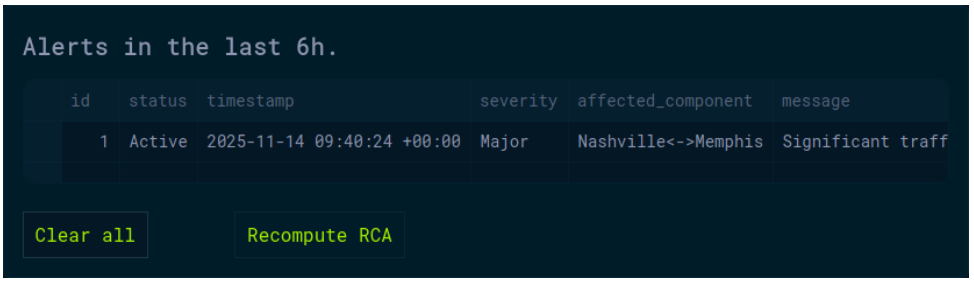
The script will introduce a change to the network that will result in traffic loss between Memphis and Nashville. However no customer is reporting any problems with any service. The task is to find what is the actual status of the link and what is broken.

Get familiar with alerting and RCA in the UI

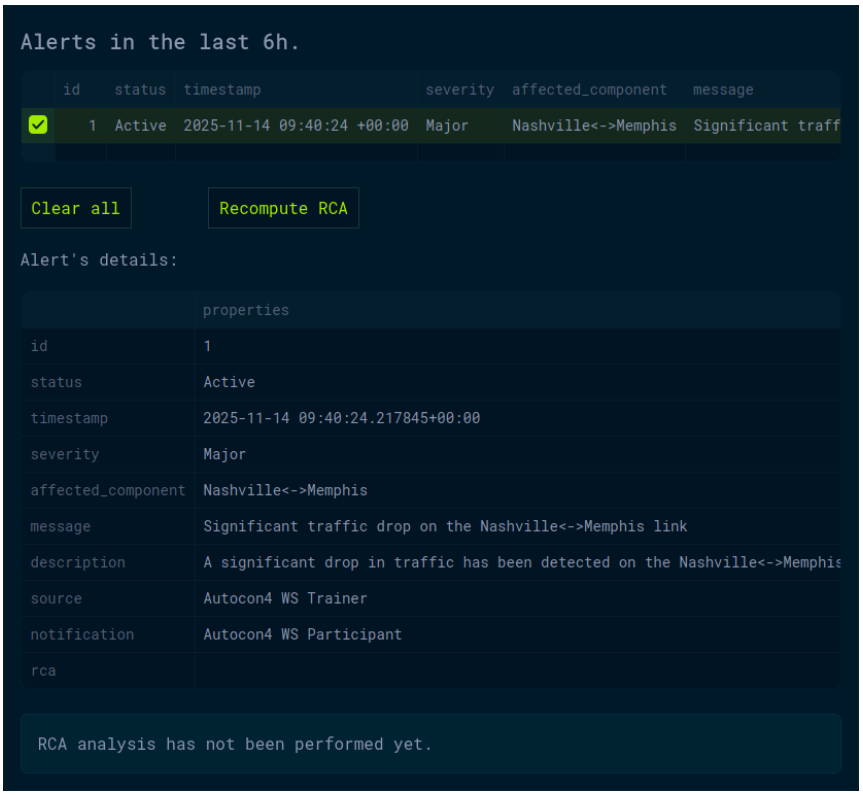
During this case an alert will be generated which will be visible in the UI. One can see that alert and read the output of the Root Cause Analysis (RCA) module and check if its findings are consistent with the investigation.

Note: In order to view alerts you must close the chat first using the “X Chat” buttton on top. And then click on “Alerts”

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Due to unresolved bug in Streamlit in order to compute RCA one has to click on “invisible checkbox” left of the Alert id:



Now one can use the “Recompute RCA” button to perform alert analysis.

Restore the network to a normal state

Run the following script:

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```
$ ~/clab/scripts/case2_fix.sh
```

Note: ***We highly recommend also clearing existing alert(s) in UI before proceeding to the next part.***

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Part 3 - High CPU utilization on P/PE router

Objectives

- Investigate higher CPU utilization on one of P/PE routers

Steps

Introduce a failure

Execute the following script in the VM shell:

```
$ ~/clab/scripts/case3_fail.sh
```

Investigate

During routine work one had discovered that one of P/PE devices had higher than usual CPU utilization in the control plane (which wasn't discovered by existing NMS). That person created a ticket (Alert) with their findings. The aim of this task is to find the underlying cause.

Note: Due to the nature of Containerlab topology, real CPU change will be very small. Scaling topology where this scenario would cause significantly higher CPU use would cause network instability due to the limited resources of VM hosting Containerlab.

Restore the network to a normal state

Run the following script:

```
$ ~/clab/scripts/case3_fix.sh
```

Note: We highly recommend also clearing existing alert(s) in UI before proceeding to the next part.

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Part 4 - Internet traffic issues towards selected IP

Objectives

- Find the underlying reason why users in selected cities cannot access site: `http://172.30.245.254`

Steps

Introduce a failure

Execute the following script in the VM shell:

```
$ ~/clab/scripts/case4_fail.sh
```

Investigate

The script will introduce a change to the network that will result in such a setup where users from selected cities (some of those cities (but not) all will be mentioned in the alert) cannot access the site located at `http://172.30.245.254` . Such users will be getting HTTP 403 errors. Other users will be able to access the site just fine. The task is to find the underlying cause.

Note: Contrary to first thought, this issue is not caused by bad nginx configuration, it's still a network issue. No nginx (web) debugging skills are needed.

Restore the network to a normal state

Run the following script:

```
$ ~/clab/scripts/case4_fix.sh
```

Note: We highly recommend also clearing existing alert(s) in UI before proceeding to the next part.

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[OPTIONAL] Part 5 - Working on your own

If there is some time left or if you finish before others, feel free to play with the topology and AI. Please note that the whole topology will be destroyed shortly after the workshop.

You have completed the hands-on successfully. Thank you for your commitment 🙂

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